Building types

unreal_kit.modern · 27 types · Runtime Toolsets, Unreal Engine 5.8 · 2026-10-27

The Python kit builds **27 kinds of contemporary building** in a running game, each with one call: homes, offices and towers, schools, a hospital, places of worship, shops, a petrol station, a factory. Every choice comes from a seed (shape, facade, colours, proportions, roof and the type's own site elements), so a street of them never repeats, and the same seed always gives the same building. Storey heights and proportions follow real buildings: 3.8 m classroom storeys, dock doors at trailer height, car-park decks about 3 m apart.

Make one

```
from unreal_kit import modern
modern.place("hospital", at=(0, 0, 0), seed=7)   # builds it in the game, returns its scene node
modern.place("apartment", at=(5000, 0, 0), yaw=90, floors=12, skin="balconies")
parts = modern.building("office_tower", seed=4)  # only the parts list, nothing built yet
```

Option	What it does
kind	One of the types in this document.
seed	Same seed, same building; a new seed gives a new design of the same type.
floors / width / depth	Exact storeys and lot size in cm (the type's range otherwise).
polygon	Build on any footprint, e.g. a GIS plot: world cm, the parts come back in world coordinates.
form / skin	Force a composition (forms below) or a facade system (facades below).
detail	high, medium or low: fewer parts for distant buildings.
site	False builds the building alone, without its plaza, parking, sports field or other site elements.
at / yaw (place)	Where the lot's front centre goes and which way the front faces (front faces -y at yaw 0).

How a building is made at runtime

1. **Design, on the agent's side.** modern.building runs in Python, outside the engine: from the seed it picks a form, outlines each volume (rectangle, rounded, chamfered, ellipse, curved bar or the given polygon), wraps it in a facade system made of whole-height or whole-perimeter pieces (no per-window modules), adds the ground floor, roof, crown and the type's site elements, and returns a list of parts: shape, size, position, material.
2. **Check.** uk.place_many checks every part first, so a mistake fails before anything is built.
3. **Send.** The parts go to the game as build.create_assembly calls inside ops.run_batch: up to 12,000 parts per request and three requests in flight. A typical building (22 to 816 parts here) is one request.
4. **Build, in the game.** The builder turns the parts into dynamic meshes with shared materials on the game thread (about 4 ms per frame, so the game keeps rendering) and registers the building as one scene node: it can be moved, styled, saved as a template and undone like anything else.
5. **Plant.** modern.place then adds the building's trees, shrubs and benches as GPU instances of the project's real meshes (RuntimeFoliage and RuntimeArchitecture props).

In the proof run every type built in the packaged game in about a second (6 s each including a 5 s wait for the screenshot). Districts (unreal_kit.district) and GIS plots (gis.set_plot building) use the same types.

Forms and facades

Each type draws from its own short list of these:

Form	Shape
block	one box
bar	a long slab
L	L-shaped
U	U-shaped
stepped	terraced steps
cantilever	an upper part overhanging
notched	a box with a cut corner
curve	a curved bar
courtyard	around a courtyard
podium_tower	a tower on a podium
setback	upper floors set back
round	round or elliptical
wedge	a wedge
H	H-shaped
bridge	two blocks joined by a bridge
twin	two towers
tower	a slim tower
campus	several linked wings
mall	anchor blocks around a concourse

Facade	What it looks like
balconies	continuous balconies with railings
punched	punched windows in solid walls
screen	a timber or metal screen
ribbon	ribbon windows between solid bands
curtain	glass curtain wall with mullions
grid	a structural grid (exoskeleton) over glass
louvres	horizontal sun louvres
tiles	tiled or panel cladding
monolith	mostly solid walls, few openings
fins	vertical fins over glass
diagrid	a diagonal grid
ribbed	ribbed metal cladding
open_deck	open car-park decks

Homes



Apartment modern.place("apartment")

Floors 5 to 14 · **Lot** 24 m to 42 m wide, 16 m to 28 m deep

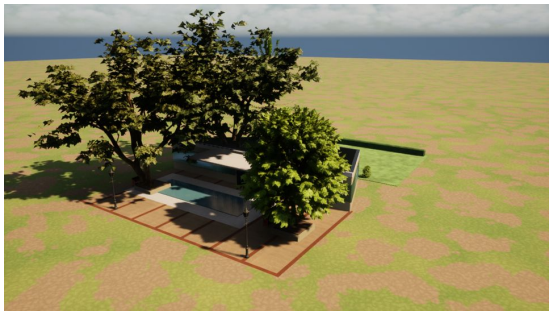
Forms block, bar, L, U, stepped, cantilever, notched, curve, courtyard, podium_tower, setback, round, wedge

Facades balconies, punched, screen, ribbon, curtain, grid, louvres, tiles

Roofs plant, green, terrace, overhang, pv

Around it Entrance canopy.

The pictured example: 100 parts, 34 x 50 x 36 m



Villa modern.place("villa")

Floors 1 to 3 · **Lot** 18 m to 26 m wide, 16 m to 24 m deep

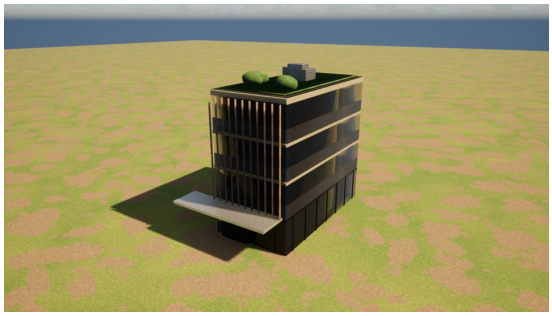
Forms cantilever, L, block, stepped, wedge

Facades punched, ribbon, screen, monolith, curtain

Roofs overhang, green, terrace, plant

Around it Swimming pool on a stone, timber or white deck; garden trees; small entrance canopy.

The pictured example: 45 parts, 26 x 34 x 5 m



Townhouse modern.place("townhouse")

Floors 3 to 4 · **Lot** 6 m to 9 m wide, 12 m to 16 m deep

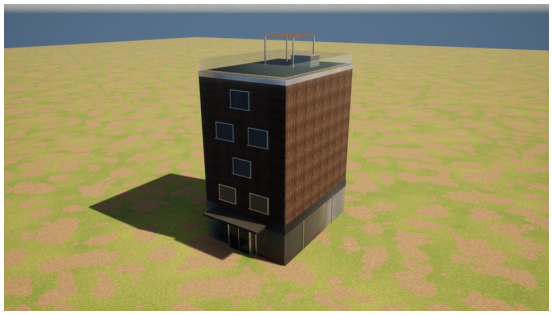
Forms block

Facades punched, screen, monolith

Roofs terrace, green, plant

Around it Built to the lot edges (party walls), so a row of them forms a terrace.

The pictured example: 65 parts, 10 x 17 x 14 m



Shophouse modern.place("shophouse")

Floors 3 to 5 · **Lot** 6 m to 10 m wide, 12 m to 18 m deep

Forms block

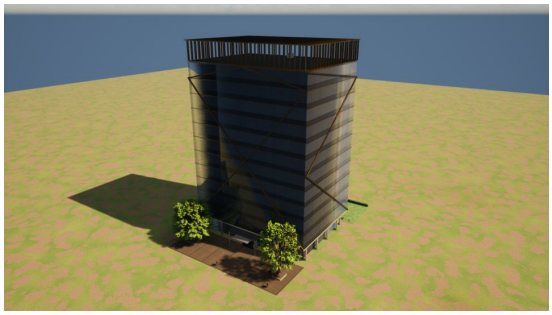
Facades punched, screen, louvres, tiles

Roofs terrace, plant

Around it Shopfront ground floor; party walls for a street of them.

The pictured example: 91 parts, 10 x 15 x 20 m

Work



Office modern.place("office")

Floors 5 to 12 · **Lot** 28 m to 48 m wide, 22 m to 36 m deep

Forms block, notched, stepped, H, L, cantilever, bridge, curve, wedge, setback, round

Facades curtain, fins, grid, louvres, ribbon, diagrid, screen, tiles

Roofs plant, pv, green, terrace · **Crowns** glow, fins_up, lantern

Around it Front plaza in granite or paving with trees (sometimes a fountain); entrance canopy.

The pictured example: 269 parts, 37 x 52 x 52 m



Office Tower modern.place("office_tower")

Floors 14 to 40 · **Lot** 30 m to 46 m wide, 30 m to 46 m deep

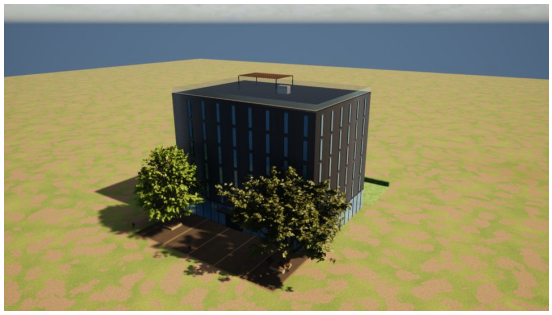
Forms podium_tower, twin, setback, tower

Facades curtain, fins, diagrid, grid, louvres

Roofs plant · **Crowns** fins_up, glow, spire, lantern, slope

Around it Front plaza; a crown on top: fins, a glowing top, a spire, a lantern or a sloped cap.

The pictured example: 231 parts, 38 x 54 x 89 m



Loft modern.place("loft")

Floors 3 to 6 · **Lot** 24 m to 40 m wide, 18 m to 30 m deep

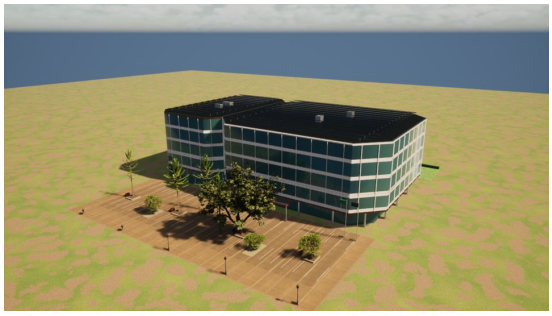
Forms block, L, courtyard, stepped, notched

Facades punched, ribbed, grid, tiles

Roofs plant, pv, terrace

Around it Brick, dark brick, charcoal or concrete cladding; no site elements.

The pictured example: 816 parts, 35 x 50 x 28 m



Business Park modern.place("business_park")

Floors 3 to 5 · **Lot** 50 m to 80 m wide, 60 m to 90 m deep

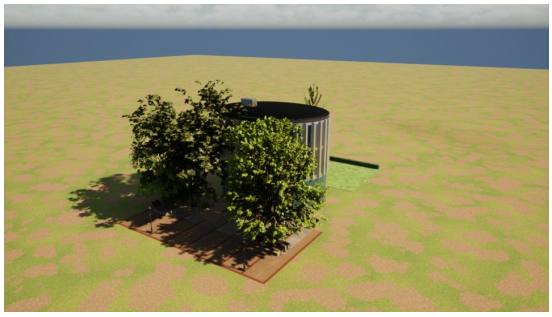
Forms H, L, U, campus, bridge, curve, cantilever, stepped

Facades curtain, fins, ribbon, louvres, grid

Roofs pv, green, plant

Around it Plaza, a parking lot in front and flagpoles.

The pictured example: 429 parts, 72 x 84 x 22 m



Bank modern.place("bank")

Floors 2 to 6 · **Lot** 22 m to 32 m wide, 20 m to 30 m deep

Forms block, notched, round, wedge

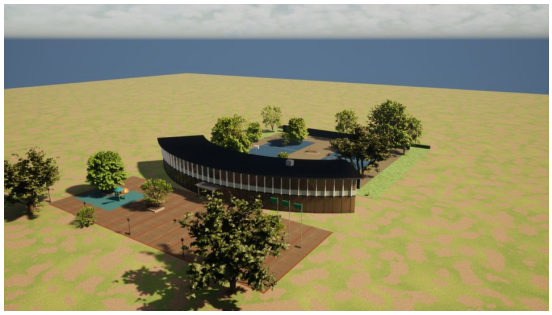
Facades monolith, grid, curtain, fins

Roofs plant

Around it Granite or paved plaza with trees; entrance canopy.

The pictured example: 125 parts, 24 x 41 x 15 m

Education, health and public service



School modern.place("school")

Floors 2 to 3 · **Lot** 60 m to 90 m wide, 70 m to 100 m deep

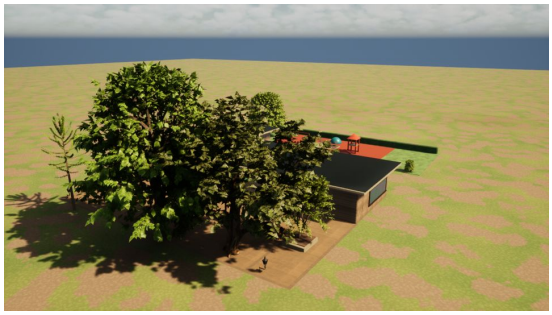
Forms L, U, courtyard, H, campus, curve, bar

Facades ribbon, punched, fins, screen, tiles

Roofs pv, green, plant, overhang

Around it Entrance canopy, flagpoles, a sports field behind the building, a playground and trees.

The pictured example: 284 parts, 66 x 102 x 10 m



Kindergarten modern.place("kindergarten")

Floors 1 to 2 · **Lot** 30 m to 45 m wide, 35 m to 50 m deep

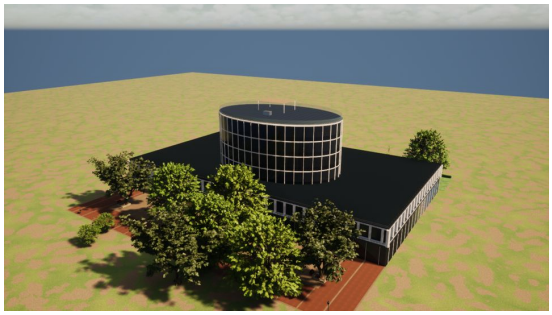
Forms L, block, campus, curve, cantilever

Facades punched, screen, tiles, ribbon

Roofs green, overhang, pv

Around it Entrance canopy, a fenced playground and trees.

The pictured example: 90 parts, 37 x 48 x 6 m



University modern.place("university")

Floors 4 to 7 · **Lot** 60 m to 90 m wide, 50 m to 80 m deep

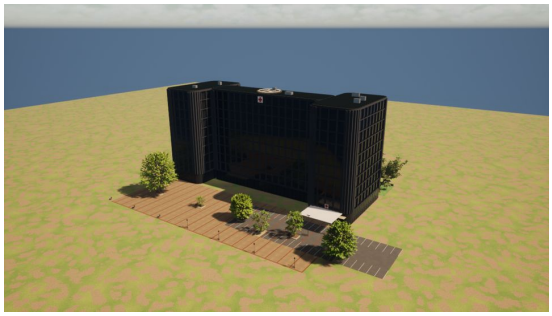
Forms courtyard, H, U, campus, bridge, curve, cantilever, podium_tower

Facades grid, fins, curtain, ribbon, screen, monolith

Roofs green, pv, plant, terrace

Around it Paved forecourt, three flagpoles and four to eight trees.

The pictured example: 338 parts, 87 x 87 x 28 m



Hospital modern.place("hospital")

Floors 6 to 12 · **Lot** 70 m to 100 m wide, 60 m to 90 m deep

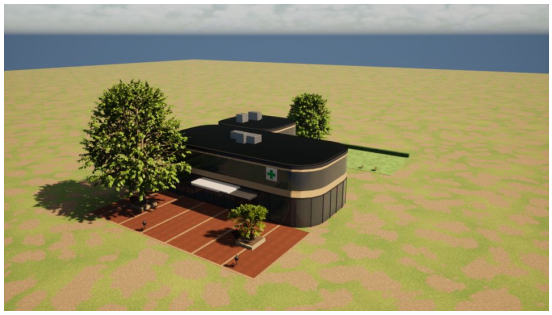
Forms podium_tower, twin, H, campus

Facades ribbon, grid, curtain, punched, fins, louvres

Roofs plant

Around it Helipad on the tallest block, main entrance canopy, emergency canopy and red or green cross signs, parking.

The pictured example: 548 parts, 115 x 83 x 45 m



Clinic modern.place("clinic")

Floors 2 to 3 · **Lot** 25 m to 40 m wide, 25 m to 35 m deep

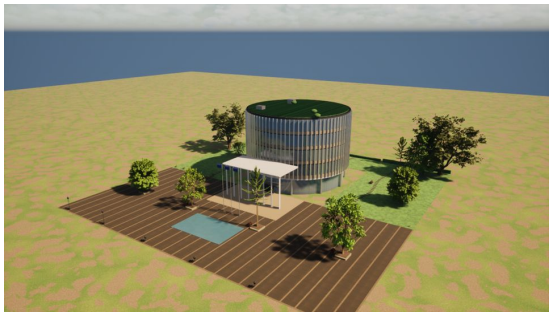
Forms block, L, notched, curve, cantilever

Facades ribbon, punched, fins, screen, curtain

Roofs plant, green, pv

Around it Entrance canopy and a cross sign.

The pictured example: 116 parts, 32 x 53 x 10 m



Government modern.place("government")

Floors 3 to 5 · **Lot** 60 m to 90 m wide, 50 m to 70 m deep

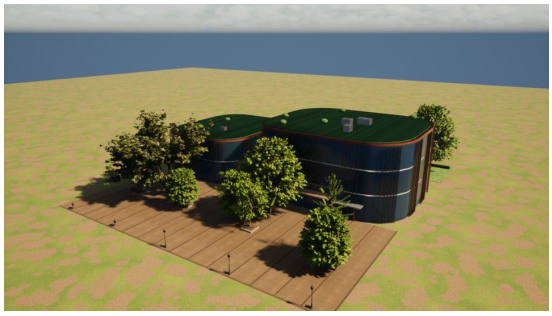
Forms block, U, H, courtyard, round

Facades grid, fins, monolith, curtain

Roofs plant, green

Around it A big thin roof on slender columns, a stone forecourt, a reflecting pool and three tall flagpoles.

The pictured example: 306 parts, 82 x 86 x 24 m



Museum modern.place("museum")

Floors 2 to 4 · **Lot** 50 m to 80 m wide, 40 m to 60 m deep

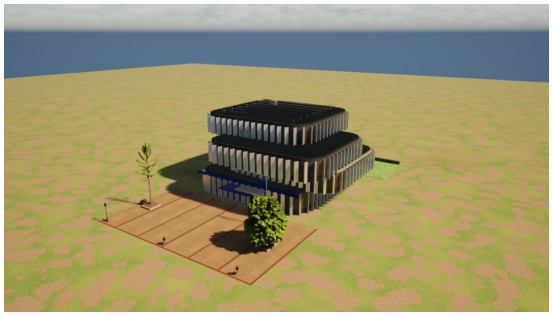
Forms cantilever, wedge, round, stepped, curve, notched

Facades monolith, tiles, screen, curtain

Roofs plant, green, overhang

Around it Granite or paved plaza with trees.

The pictured example: 291 parts, 70 x 76 x 18 m



Police Station modern.place("police_station")

Floors 2 to 4 · **Lot** 30 m to 50 m wide, 30 m to 45 m deep

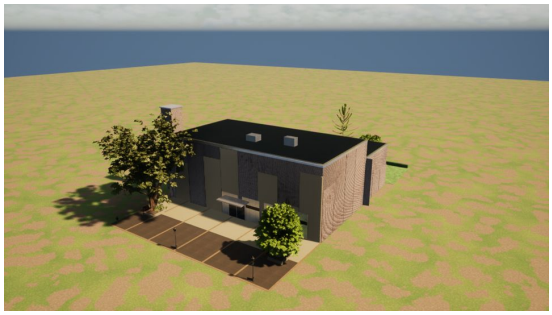
Forms block, L, notched, cantilever

Facades punched, ribbon, grid, fins

Roofs plant, pv

Around it Blue entrance canopy and fascia band; a flagpole.

The pictured example: 308 parts, 37 x 63 x 17 m



Fire Station modern.place("fire_station")

Floors 2 to 3 · **Lot** 32 m to 45 m wide, 26 m to 40 m deep

Forms block, L

Facades punched, ribbed, tiles, ribbon

Roofs plant, pv

Around it Engine bay doors (red, white or aluminium) with glazing, a concrete apron and a hose-drying tower.

The pictured example: 90 parts, 40 x 61 x 18 m

Faith



Mosque modern.place("mosque")

Floors 1 · **Lot** 26 m to 42 m wide, 26 m to 42 m deep

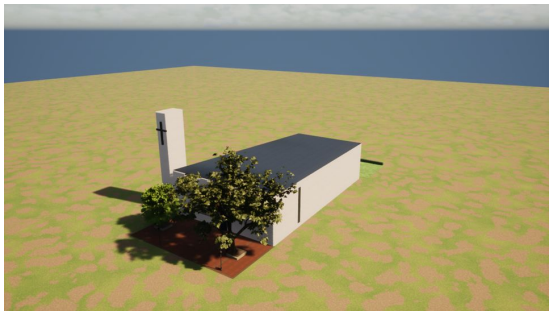
Forms block

Facades monolith, punched

Roofs flat

Around it A dome (hemisphere, pointed or onion; gold, white, verdigris, blue or copper), sometimes four small domes; one, two or four minarets; a marble or paved forecourt.

The pictured example: 103 parts, 37 x 48 x 21 m



Church modern.place("church")

Floors 1 · **Lot** 20 m to 28 m wide, 35 m to 50 m deep

Forms block, wedge

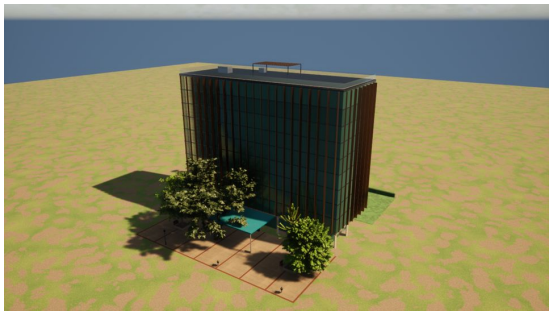
Facades monolith, screen

Roofs overhang

Around it A bell tower with a cross beside the nave.

The pictured example: 36 parts, 32 x 69 x 22 m

Shops, hotels and fuel



Hotel modern.place("hotel")

Floors 8 to 20 · **Lot** 35 m to 50 m wide, 35 m to 50 m deep

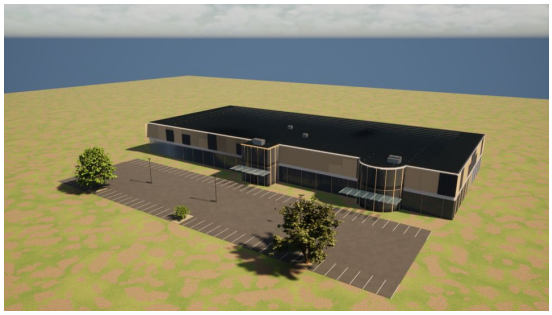
Forms podium_tower, bar, curve, setback, tower, L

Facades balconies, fins, curtain, louvres, screen, punched

Roofs terrace, plant · **Crowns** glow, lantern, fins_up

Around it Entrance canopy over a granite drop-off, a rooftop pool on a timber deck, trees.

The pictured example: 197 parts, 45 x 53 x 38 m



Mall modern.place("mall")

Floors 2 to 3 · **Lot** 90 m to 140 m wide, 80 m to 120 m deep

Forms mall

Facades tiles, monolith, ribbed

Roofs plant, pv

Around it Entry canopies on each entrance block and a large parking lot.

The pictured example: 377 parts, 111 x 89 x 13 m



Supermarket modern.place("supermarket")

Floors 1 · **Lot** 40 m to 60 m wide, 50 m to 70 m deep

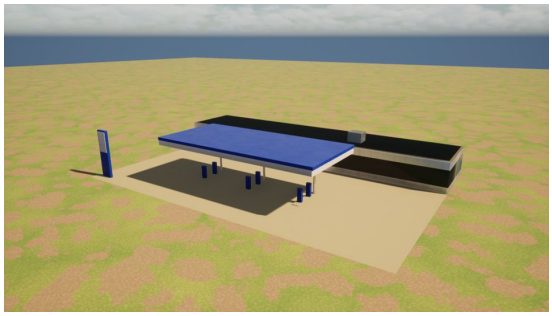
Forms block

Facades tiles, ribbed, monolith

Roofs pv, plant

Around it Entry canopy and a parking lot.

The pictured example: 91 parts, 52 x 57 x 8 m



Petrol Station modern.place("petrol_station")

Floors 1 · **Lot** 35 m to 50 m wide, 35 m to 50 m deep

Forms block

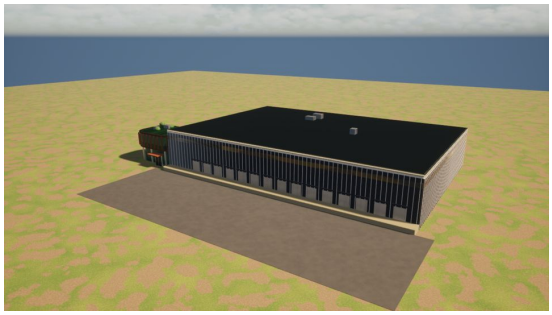
Facades tiles, ribbon, curtain

Roofs plant

Around it Forecourt, a canopy over two or more pump islands and a price pylon, all in one brand colour.

The pictured example: 22 parts, 45 x 31 x 7 m

Industry and parking



Warehouse modern.place("warehouse")

Floors 1 · **Lot** 60 m to 120 m wide, 50 m to 90 m deep

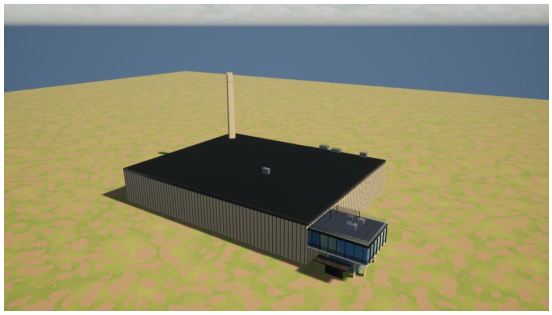
Forms block

Facades ribbed, tiles

Roofs pv, plant

Around it Loading dock doors with canopies, a dock apron, a truck yard and an office annex.

The pictured example: 330 parts, 90 x 76 x 13 m



Factory modern.place("factory")

Floors 1 · **Lot** 60 m to 100 m wide, 50 m to 80 m deep

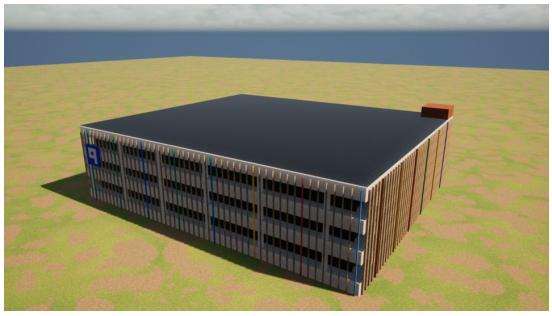
Forms block, L

Facades ribbed, tiles

Roofs barrel, pv, plant

Around it Office annex, one to three chimneys and up to four silos.

The pictured example: 352 parts, 78 x 55 x 35 m



Parking Garage modern.place("parking_garage")

Floors 3 to 6 · **Lot** 35 m to 50 m wide, 30 m to 45 m deep

Forms block

Facades open_deck

Roofs flat

Around it Open parking decks, a blue P sign and a stair core.

The pictured example: 230 parts, 44 x 38 x 12 m